HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2025)

Student No.: 03 Name: _____ Marks: / 115

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If a and b are constants such that $(x+8)(x+a)+b \equiv x^2+5a(x+3)$, then $b=$

   A. -14.

   B. -2.

   C. 2.

   D. 14.

(DSE-2025 PII Q03)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

2. If $(3c+1)(d-4) = 2d(5c-1)$, then $c =$

A. $\dfrac{3d-4}{7d+12}$.

B. $\dfrac{3d+4}{7d-12}$.

C. $\dfrac{7d-12}{3d+4}$.

D. $\dfrac{7d+12}{3d-4}$.

(DSE-2025 PII Q04)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

3. Let a_n be the nth term of a sequence. If $a_2 = 3$, $a_5 = 41$ and $a_{n+2} = 2a_{n+1} + a_n$ for any positive integer n, then $a_6 =$

A. 99.

B. 101.

C. 239.

D. 243.

(DSE-2025 PII Q13)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

4. The perimeter of the sector OMN is 12π cm, where O is the centre of the sector OMN. If the radius of the sector OMN is 3π cm, which of the following are true?

I. The area of the sector OMN is $9\pi^2$ cm^2.

II. The perimeter of $\triangle OMN$ is less than 35 cm.

III. The angle of the sector OMN is greater than $100°$.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

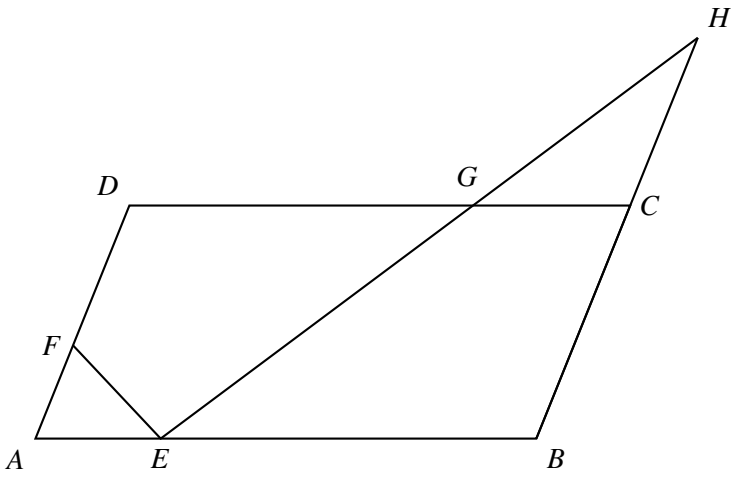
(DSE-2025 PII Q15)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

5. In the figure, $ABCD$ is a parallelogram. Let E, F and G be points lying on AB, AD and CD respectively such that $BE = 3AE$, $2DF = 3AF$ and $DG = 2CG$. It is given that BC produced and EG produced meet at the point H. If the area of $\triangle CGH$ is $16\,\text{cm}^2$, then the area of the quadrilateral $DFEG$ is



A. $46\,\text{cm}^2$.

B. $49\,\text{cm}^2$.

C. $105\,\text{cm}^2$.

D. $115\,\text{cm}^2$.

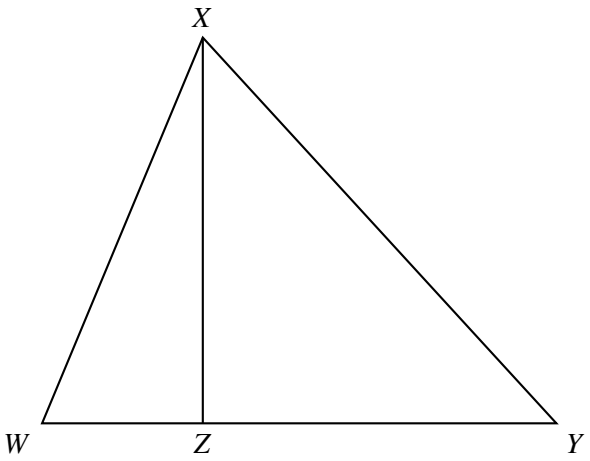
(DSE-2025 PII Q17)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

6. The figure shows the triangle WXY. Let Z be a point lying on WY such that $WY : XY = XY : YZ$. If $WX = 65\,\text{cm}$, $WZ = 25\,\text{cm}$ and $XZ = 60\,\text{cm}$, then $XY =$



A. 131 cm.

B. 144 cm.

C. 156 cm.

D. 169 cm.

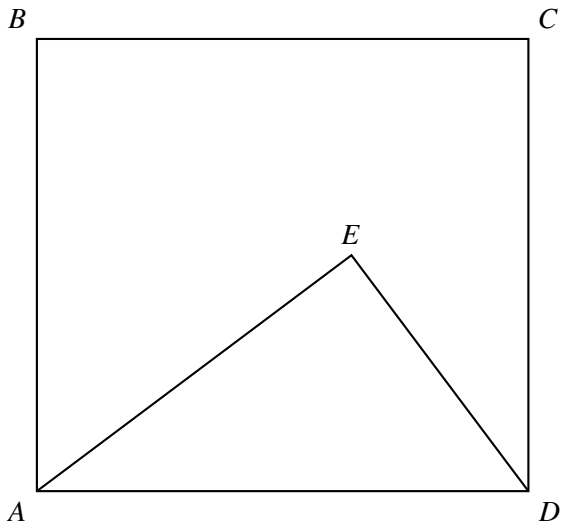
(DSE-2025 PII Q18)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

7. In the figure, $ABCD$ is a rectangle. It is given that $AE = 20\,\text{cm}$ and $\angle AED = 90°$. If the area of $\triangle ADE$ is $150\,\text{cm}^2$, then the perpendicular distance from E to CD is



A. 8 cm.

B. 9 cm.

C. 12 cm.

D. 15 cm.

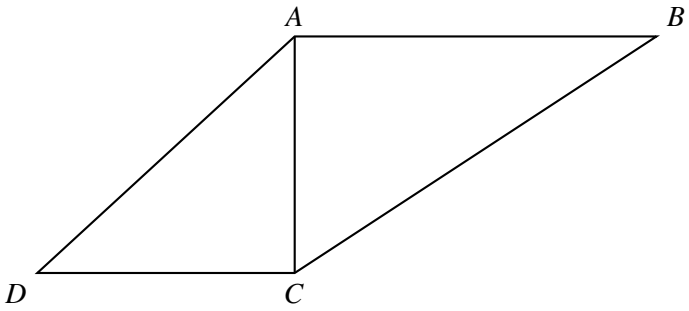
(DSE-2025 PII Q21)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

8. The figure shows the quadrilateral $ABCD$. If $\angle ABC + \angle ADC = \angle ACD = \angle BAC = 90°$, which of the following must be true?



A. $\tan\angle ACB = \dfrac{AB}{AD}$

B. $\tan\angle ACB = \dfrac{AB}{CD}$

C. $\tan\angle ADC = \dfrac{BC}{AD}$

D. $\tan\angle ADC = \dfrac{BC}{CD}$

(DSE-2025 PII Q23)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

9. The polar coordinates of the points X, Y and Z are $(1, 20^\circ)$, $(2, 80^\circ)$ and (r, θ) respectively, where $20^\circ < \theta < 80^\circ$. If $\triangle XYZ$ is an equilateral triangle, find r.

A. $\sqrt{3}$

B. $\sqrt{5}$

C. $\sqrt{7}$

D. $\sqrt{10}$

(DSE-2025 PII Q24)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

10. The straight lines L_1 and L_2 are perpendicular to each other. The equations of L_1 and L_2 are $3x+4y-20=0$ and $mx+ny-20=0$ respectively, where m and n are constants. Suppose that L_1 and L_2 cut the x-axis at the points A and B respectively. Let C be the point of intersection of L_1 and L_2. It is given that C lies above the x-axis. If the area of $\triangle ABC$ is 6, find n.

A. -12

B. -9

C. 9

D. 12

(DSE-2025 PII Q26)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

11. The coordinates of the centre of the circle C are $(7,-5)$. If C cuts the x-axis at the points P and Q such that $PQ = 24$, then the equation of C is

A. $x^2+y^2-14x+10y-95=0$.

B. $x^2+y^2+14x-10y-95=0$.

C. $x^2+y^2-14x+10y-119=0$.

D. $x^2+y^2+14x-10y-119=0$.

(DSE-2025 PII Q27)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

12. A box contains six cards numbered 2, 2, 2, 3, 4 and 5 respectively. In a game, a number is randomly drawn from the box and a certain number of tokens will be got according to the following table:

Number drawn	2	3	4	5
Number of tokens got	10	15	25	50

Find the expected number of tokens got in the game.

A. 3

B. 4

C. 20

D. 25

(DSE-2025 PII Q28)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

13. Consider the following data:

$$\alpha \quad \beta \quad -4 \quad -3 \quad 1 \quad 1 \quad 1 \quad 4$$

Denote the mode and the median of the above data by s and t respectively. If the mean and the range of the above data are 0 and 10 respectively, which of the following are true?

I. $s = 1$

II. $t = -1$

III. $\alpha + \beta = 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2025 PII Q30)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

14. The L.C.M. of p^2-4q^2, p^3-8q^3 and $(p+2q)(p^2-4q^2)$ is

A. $p-2q$.

B. p^2-4q^2.

C. $(p+2q)^2(p^3-8q^3)$.

D. $(p+2q)(p^2-4q^2)(p^3-8q^3)$.

(DSE-2025 PII Q32)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

15. It is given that $\log_5 y$ is a linear function of $\log_{25} x$. The intercepts on the vertical axis and on the horizontal axis of the graph of the linear function are 12 and 2 respectively. If $y = mx^n$, then $n =$

A. -12.

B. -4.

C. -3.

D. -1.

(DSE-2025 PII Q33)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

16. $i^9 + i^{10} + i^{11} + \cdots + i^{999} =$

A. -1.

B. 0.

C. 1.

D. i.

(DSE-2025 PII Q35)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

17. Consider the following system of inequalities:

$$\begin{cases} x \leq 11 \\ 4x + 5y - 19 \geq 0 \\ 7x - 6y + 11 \leq 0 \end{cases}$$

Let D be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in D, then the greatest value of $8x - 6y + 11$ is

A. 1.

B. 11.

C. 15.

D. 129.

(DSE-2025 PII Q36)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

18. Let p, q and r be non-zero real numbers. If p, q, r is an arithmetic sequence, which of the following must be true?

I. 3^p, 3^q, 3^r is a geometric sequence.

II. $\frac{5}{p}$, $\frac{5}{q}$, $\frac{5}{r}$ is a geometric sequence.

III. $p-q$, $q-r$, $r-p$ is an arithmetic sequence.

A. I only

B. II only

C. I and III only

D. II and III only

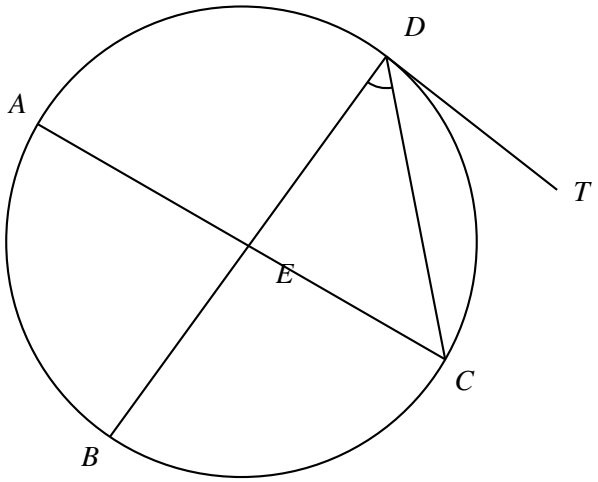
(DSE-2025 PII Q37)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

19. In the figure, AC is a diameter of the circle $ABCD$. Denote the point of intersection of AC and BD by E. It is given that TD is the tangent to the circle at D. If $\angle BEC = 96^\circ$ and $\angle CDT = 41^\circ$, then $\angle CDE =$



A. 47°.

B. 48°.

C. 52°.

D. 55°.

(DSE-2025 PII Q38)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

20. For $90° < \theta < 270°$, how many roots does the equation $\tan^3\theta = 2\tan\theta$ have?

A. 2

B. 3

C. 4

D. 5

(DSE-2025 PII Q39)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

21. $PQRS$ is a regular tetrahedron. Find the angle between PQ and $\triangle QRS$ correct to the nearest degree.

A. 35°

B. 55°

C. 60°

D. 71°

(DSE-2025 PII Q40)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

22. Denote the origin by O. The coordinates of the point U are $(20, 0)$. Let V be a point lying on the positive y-axis such that the x-coordinate of the in-centre of $\triangle OUV$ is 6. Find the area of $\triangle OUV$.

A. 70

B. 87

C. 210

D. 250

(DSE-2025 PII Q41)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

23. The standard scores of a boy and a girl in a Mathematics test are -2 and z respectively. The standard deviation of the scores of the Mathematics test is 2 marks. If the difference of the test score of the boy and the test score of the girl is 6 marks, find z.

A. -5 or 1

B. -5 or 3

C. -3 or 1

D. -3 or 3

(DSE-2025 PII Q44)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

24. Consider the compound inequality

$$\frac{6x+1}{2} < x-8 \quad \text{or} \quad 3x \leq -21 \quad \cdots\cdots\cdots\cdots (*).$$

(a) Solve $(*)$.

(b) Write down the greatest integer satisfying $(*)$.

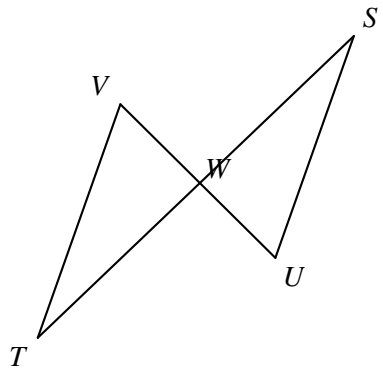
(4 marks) (DSE-2025 PI Q06)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

25. In the figure, ST and UV intersect at the point W. It is given that $SU \parallel VT$ and W is the mid-point of ST.



(a) Prove that $\triangle SUW \cong \triangle TVW$.

(b) Let X be a point lying on TW such that $\triangle SUW \sim \triangle VXW$. If $SU = 57\,\text{cm}$, $SW = 63\,\text{cm}$ and $WX = 7\,\text{cm}$, find the perimeter of $\triangle TVX$.

(5 marks) (DSE-2025 PI Q08)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

26. It is given that $p(x)$ partly varies as x and partly varies as x^2. Suppose that $p(7) = 56$ and $p(9) = 216$.

(a) Find $p(x)$.

(b) Let c be a real constant. Find the range of values of c such that the equation $p(x) = c$ has two distinct real roots.

(6 marks) (DSE-2025 PI Q11)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

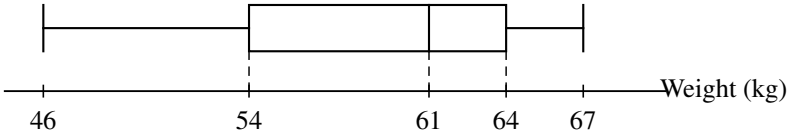
27. The stem-and-leaf diagram below shows the distribution of the weights (in kg) of some athletes before training.

Stem (tens)	Leaf (units)
4	5 6 8
5	w w 6 7
6	3 4 5 6 9 9
7	0 2
8	7

The difference of the range and the inter-quartile range of the above distribution is 25 kg.

(a) Write down the range of the above distribution. Hence, find w.

(b) The box-and-whisker diagram below shows the distribution of the weights (in kg) of the athletes after the training.



(i) Find the change in the upper quartile of the distribution due to the training.

(ii) Is the distribution of the weights of the athletes after the training less dispersed than that before the training? Explain your answer.

(7 marks) (DSE-2025 PI Q12)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

28. The straight line L cuts the positive x-axis and the positive y-axis at the points M and N respectively. Let P be a moving point in the rectangular coordinate plane such that $MP = NP$. Denote the locus of P by Γ. It is given that the equation of Γ is $3x - 2y - 30 = 0$.

(a) Describe the geometric relationship between Γ and MN.

(b) Find the equation of L.

(c) Suppose that Γ cuts L and the x-axis at the points Q and R respectively. Find the area of the quadrilateral $ONQR$, where O is the origin.

(7 marks) (DSE-2025 PI Q13)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

29. The height and the base radius of a solid metal right circular cone are 45 cm and 24 cm respectively. The circular cone is divided into a circular cone and a frustum X by a plane parallel to its base. It is given that the height of X is 30 cm.

(a) Express the volume of X in terms of π.

(b) Express the total surface area of X in terms of π.

(c) X is melted and recast into a solid cube. Does the total surface area of the cube exceed the total surface area of X? Explain your answer.

(9 marks) (DSE-2025 PI Q14)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

30. It is given that $\begin{cases} \log_3 x + \log_3 y = 9 \\ \log_x 81 - \log_y 9 = 1 \end{cases}$, where $0 < x < y$.

(a) Let $u = \log_3 y$. Prove that $u^2 - 3u - 18 = 0$.

(b) Find x.

(4 marks) (DSE-2025 PI Q16)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

31. Let $T(n)$ be the nth term of an arithmetic sequence such that $T(1) \neq T(2)$ and $T(47) = 456$. It is given that $T(9)$, $T(47)$, $T(199)$ is a geometric sequence.

(a) Find $T(1)$.

(b) Find the least value of n such that the sum of the first n terms of the arithmetic sequence is greater than 10^6.

(6 marks) (DSE-2025 PI Q17)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

32. Let $g(x) = 3x^2 - 6kx + 24x + 3k^2 - 24k + 55$, where k is a positive constant. Denote the vertex of the graph of $y = g(x)$ by R.

(a) Using the method of completing the square, express the coordinates of R in terms of k.

(b) The graph of $y = h(x)$ is obtained by translating the graph of $y = g(x)$ rightwards by 6 units and then downwards by 10 units. Denote the vertex of the graph of $y = h(x)$ by S. Let T be a point such that the orthocentre of $\triangle RST$ is the origin.

(i) Express the coordinates of T in terms of k.

(ii) The coordinates of the point U are $(-5, 5)$. Let V be a point lying on RS such that TV is perpendicular to RS. It is given that S, T, U and V are concyclic. Find k.

(9 marks) (DSE-2025 PI Q18)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

33. The equation of the straight line L is $4x-3y+83=0$. The equation of the circle C is $x^2+y^2+ax-2y+b=0$, where a and b are positive constants. It is given that C passes through the point $(-10,9)$ and L is a tangent to C.

(a) Prove that the equation $25x^2+(9a+640)x+(90a+4924)=0$ has equal roots. Hence, find a and b.

(b) The coordinates of the point P are $(-20,1)$. Let Q be a point lying below L such that PQ is tangent to C and $PQ=25$. Denote the centre of C by I.

(i) Find $\cos\angle IPQ$.

(ii) Find, in surd form, the distance between I and Q.

(iii) Let R be a point lying on L such that C is the inscribed circle of $\triangle PQR$. Denote the radius of the circumcircle of $\triangle PQR$ by r. Someone claims that $r>PQ$. Is the claim correct? Explain your answer.

(12 marks) (DSE-2025 PI Q19)

I can finish _____ questions

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.